Ocean Moored Buoy Layout Optimization

***Abstract*—With the intensification of global climate change, the frequent occurrence of extreme disasters such as tropical cyclones and submarine earthquakes poses a severe threat to human society and marine ecosystems. Developing effective ocean observation networks to enhance disaster warning capabilities has become an urgent necessity. As noted in *Nature*, climate scientist Roxy Mathew Koll from the Indian Institute of Tropical Meteorology stated that buoy monitoring data is crucial for predicting tropical cyclones, as some region accounts for over 80% of global fatalities from such events. Research indicates that optimizing the layout of moored ocean buoy is key to enhancing their monitoring value. However, buoy layout optimization is a classic high-dimensional, multimodal, and strongly constrained multi-objective optimization problem. The core challenge lies in maximizing monitoring coverage of high-risk disaster areas while satisfying a variety of complex physical and geographical constraints. To address this challenge, this paper proposes a data-driven optimization framework that integrates four key models: 1) a seismic hazard probability model based on historical data; 2) a tropical cyclone hazard probability model, also based on historical data, with both constructed using Kernel Density Estimation (KDE); 3) a buoy spatial influence and coverage density model to ensure uniform coverage; and 4) a geospatial constraint model incorporating land elevation, sea depth, and coastal proximity. Due to the complexity of this optimization model, traditional mathematical programming methods are inadequate for finding a solution. This study demonstrates that the proposed model, when combined with the historical archive-assisted constrained multimodal multi-objective optimization algorithm (HACMMO), can effectively solve the moored ocean buoy layout problem. This provides a scientific decision-making tool for enhancing global ocean environmental monitoring and disaster warning capabilities. The proposed HACMMO optimization algorithm demonstrated superior performance by achieving the best results in 8 out of 9 test scenarios compared to other state-of-the-art (SOTA) algorithms. This enhancement leads to a more robust and effective buoy layout, which is critical for monitoring disasters like tropical cyclone events and submarine earthquakes. This advanced tool, therefore, realizes significant and quantifiable environmental and societal benefits.**

***Index Terms*—Ocean Moored Buoy; Site Selection Optimization; Constrained Multimodal Multi-Objective Optimization; Ocean Climate Monitoring**

I. Introduction

The ocean plays a vital role in the global climate system and the sustainable development of human society. However, the increasing frequency and rapid intensification of extreme weather events and geological hazards poses a significant threat to coastal regions and global marine activities [1]. Therefore, establishing a long-term, continuous, and stable ocean observation system is indispensable for disaster warning, scientific research, and the protection of the Earth's ecosystem [2].

Moored ocean buoy, as key in-situ observation platforms, can be equipped with various sensors to collect and transmit critical hydrographic and meteorological data in real time [3]. They form the core component of modern ocean observation networks. Internationally, several large-scale buoy observation networks have been successfully deployed, such as the Global Ocean Observing System coordinated by the World Meteorological Organization , which includes the renowned Argo program [4], TAO/TRITON array [5], PIRATA array [6], and RAMA array [7]. These networks have played a crucial role in monitoring large-scale ocean-atmosphere interaction phenomena like the El Niño-Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD) [8].

Despite significant progress in the construction of global buoy networks, the optimization of their spatial layout remains a formidable scientific challenge. Current layout designs often rely on historical project experience and specific scientific objectives, lacking a systematic, data-driven top-level design framework. This can lead to redundant monitoring in some sea areas while leaving other critical, high-risk regions with observational gaps.

From a mathematical perspective, buoy layout optimization is a complex multi-objective problem. However, applying traditional facility location models, such as the p-Median Problem and the Maximal Covering Location Problem [9, 10], lead to significant discrepancies. The core assumption of these models is that demand points originate from a predefined, discrete set of geographical points. In the real-world scenario of marine disaster warning, demand points are not a set of fixed cities or customer locations but rather a continuous and dynamic probabilistic risk field, determined by the spatiotemporal distribution of hazards like tropical cyclones and submarine earthquakes. This risk distribution, driven by real historical data, is highly non-linear, non-differentiable, and non-convex, rendering it a black-box model that cannot be described by a simple analytical expression.

Furthermore, the problem must be solved within a vast continuous geographical space, rather than a finite set of candidate sites, while satisfying a series of complex and strict physical and geospatial constraints.

Faced with this constrained multimodal multi-objective optimization problem (CMMOP), which is characterized by a data-driven black-box probability model, a continuous solution space, and complex real-world constraints, traditional mathematical programming algorithms are no longer applicable. Concurrently, while existing intelligent optimization algorithms have been applied to similar spatial optimization problems [11-13], their performance and robustness are challenged by such complex real-world constraints and multimodal characteristics, facing the risk of becoming trapped in local optima.

To address these challenges, this paper aims to develop a comprehensive framework for optimizing the layout of global moored ocean buoy. The main contributions of this work are as follows:

(1) Construction of a Multi-Objective Model Integrating Multi-Source Hazard Data. This model, for the first time, incorporates probability density distributions of historical earthquakes and tropical cyclones as core optimization

objectives. This prioritizes buoy placement in high-risk areas to maximize their early warning value.

(2) Systematic Definition of Physical and Geospatial Constraints. The framework not only considers suitable water depth to ensure equipment safety but also introduces a minimum distance-from-land constraint to guarantee the representativeness of observational data, thereby making the optimized results more practical.

(3) Validation of an advanced optimization algorithm. To address the multimodal, multi-objective, and complex constraint characteristics of the model, we apply the HACMMO algorithm. We demonstrate its effectiveness in providing high-quality solutions for this large-scale spatial optimization problem.

The remainder of this paper is organized as follows: Section II introduces the related work of site selection optimization. Section III provides a detailed description of the problem modeling process, including the datasets, optimization objectives, the four key sub-models, and the proof of the model's non-linear, non-differentiable, and non-convex. Section IV introduces the HACMMO algorithm used to solve this problem. Section V presents the experimental design and result analysis, validating the proposed method's effectiveness. Finally, Section VI concludes the paper and discusses future research directions.

II. Related Work

The optimization of moored ocean buoy layouts is, in essence, a complex variant of the Facility Location Problem (FLP) [12]. FLP is a classic research area in operations research and geographic information science, aiming to determine the optimal geographical locations for a set of facilities to optimize one or more objectives [14].

Classic location models, such as the Maximal Covering Location Problem and the p-Median Problem [9, 10], have historically provided foundational frameworks for site selection. These models are typically solved using mathematical programming or standard heuristic algorithms like Genetic Algorithms (GAs) and Particle Swarm Optimization (PSO) [11-13]. However, these traditional approaches are all predicated on a critical assumption: the locations of demand points are discrete, predetermined, and known. This assumption creates a significant disconnect between theory and the reality of buoy layout optimization. In this real-world scenario, demand points are not a set of candidate facilities or customer locations. Instead, it is a continuous, dynamic, and probabilistic risk field defined by the complex spatiotemporal distribution of disasters like tropical cyclones and submarine earthquakes. Directly applying classic location models would oversimplify the true nature of disaster risk, failing to capture high-risk corridors between historical event points and thus inadequately reflecting the actual needs for early disaster warning.

In recent years, with the advancement of machine learning and multi-source spatial data fusion, some studies have begun to explore data-driven approaches for facility site optimization [15, 16]. These methods leverage historical data to model demand more realistically. However, research specifically targeting the global-scale moored ocean buoy layout problem remains scarce. A comprehensive framework that systematically integrates multiple hazard probability models, complex physical and geographical constraints, and the existing network layout has yet to be fully developed.

More importantly, the reliance on real-world historical data for modeling disaster probability, combined with complex geographical boundaries, results in an optimization problem characterized by inherently non-linear, non-differentiable, and non-convex, and black-box properties. As will be formally demonstrated in Section III-*G*, these characteristics render traditional mathematical programming methods intractable and serve as the primary motivation for adopting an advanced, data-driven optimization framework. This means their explicit mathematical expressions and gradient information are unobtainable, which renders traditional mathematical programming methods intractable. While intelligent optimization algorithms are often employed for such problems, they are typically applied to simplified models with explicit mathematical formulations. The unique complexity of the buoy layout problem—characterized by a data-driven black-box objective function, a vast continuous solution space, and stringent real-world constraints—challenges the performance and robustness of even advanced optimization algorithms, posing a high risk of premature convergence to local optima.

Therefore, this study aims to fill this gap. By moving away from the reliance on deterministic demand points found in traditional models, we propose a fully data-driven optimization framework. This framework is specifically designed to handle continuous space, probabilistic demand, and complex constraints, thereby providing more scientific and refined decision-making support for the design of global ocean observation networks.

III. Problem Modeling and Mathematical Formulation

This section details the modelling process for the moored ocean buoy layout optimization problem, including the datasets used, the core optimization objectives, the four key sub-models, and the proof of the model's non-linear, non-differentiable, and non-convex. The problem can be abstracted as follows: within the global ocean space, find suitable locations to deploy new buoys to maximize observational value and avoid redundant or sparse monitoring, without altering the positions of existing buoys.

A. Datasets

The following datasets were used to construct and validate the models (datasets sourced from Kaggle and National Oceanic and Atmospheric Administration (NOAA) open data [17-20]):

(1) Tropical Cyclone Data: Historical data from 1980 to 2023, including global tropical cyclone tracks and intensities [17]. This dataset contains approximately 300,000 records and is used to estimate the probability of cyclone occurrences.

(2) Earthquake Data: The global earthquake catalogue from 1965 to 2016, including epicentre locations and magnitudes. This dataset contains about 30,000 records and is used to estimate earthquake probability [18].

(3) Existing Buoy Data: Location information for existing global or regional buoy networks, used to evaluate the

supplementary value of new deployments and avoid redundancy [19].

(4) Geospatial Data: The ETOPO1 Global Relief Model [20], which provides land elevation and ocean bathymetry data for building the constraint model.

B. Model Objectives

The core optimization objectives for the moored ocean buoy layout can be summarized as:

(1) Maximize Tropical Cyclone Monitoring Probability, buoy should be placed in high-occurrence areas for tropical cyclones to capture the most valuable monitoring data.

(2) Maximize Earthquake Monitoring Probability, buoy should be located in seismically active zones to capture valuable monitoring data.

(3) Optimize Spatial Coverage and Dispersion, buoy should be spaced as far apart as possible to avoid redundant monitoring in small areas, thereby increasing the spatial coverage and informational independence of the entire network.

(4) Satisfy Deployment Feasibility Constraints, buoy placement locations must meet specific geographical and physical conditions:

a) Distance from Land, buoy should be placed in the open ocean to obtain representative data, avoiding significant influences from coastal runoff, tides, and human activities.

b) Suitable Water Depth, buoy need to be moored in sufficiently deep water to ensure the stability of the water mass, reflect long-term and large-scale oceanographic trends, and prevent the equipment from grounding or beaching.

Based on the objectives above, we constructed the following four key models.

C. Seismic Hazard Probability Model

The objective of this model is to construct a spatial probability density function, denoted as $P_{EQ}(x, y)$, based on historical global earthquake event data. This function estimates the relative probability of an earthquake occurring at any given geographical coordinate (x, y), where x represents longitude and y represents latitude. A higher output value from the model indicates a region with more frequent historical seismic activity and, consequently, a higher associated risk of future earthquakes. Given that the distribution of historical earthquakes exhibits irregular, multi-modal spatial patterns, traditional parametric probability distributions are inadequate for an accurate fit. Therefore, we employ the non-parametric KDE method to formulate this model [21]. KDE estimates the probability density function in a data-driven manner, offering significant flexibility and superior fitting capabilities for complex distributions.

Let the historical earthquake event dataset be $Dataset_{EQ}=\{\mathbf{d}_1,\mathbf{d}_2,\ldots,\mathbf{d}_n\}$, where n is the total number of historical seismic events. Each data point $\mathbf{d}_i=(x_i, y_i)$ is a two-dimensional vector representing the longitude and latitude coordinates of the epicenter of the i-th earthquake. Considering the Earth is a sphere, and the discontinuity at the $\pm 180^\circ$ longitude meridian poses a boundary problem for standard Euclidean-based KDE algorithms. To address this, we perform a data augmentation step. For earthquake events located near the eastern or western boundaries, we generate "virtual" copies by shifting their longitudes by 360°. The augmented dataset, $Dataset'_{EQ}$, is defined as:

$$\begin{aligned} Dataset'_{EQ} = Dataset_{EQ} \\ \cup\{(x_i-360, y_i) \mid (x_i, y_i) \\ \in Dataset_{EQ} \wedge x_i > 180-\Delta_x\} \quad , \\ \cup\{(x_i+360, y_i) \mid (x_i, y_i) \\ \in Dataset_{EQ} \wedge x_i < -180+\Delta_x\} \end{aligned} \tag{1}$$

where Δ_x is a predefined longitude buffer width (e.g., 45°). This process ensures a smooth and continuous probability density transition across the dateline.

Using the augmented dataset $Dataset'_{EQ}$ (of size n'), the seismic probability density $P_{EQ}(\mathbf{P})$ at any arbitrary location $\mathbf{P}=(x, y)$ is calculated using Gaussian KDE, expressed as:

$$P_{EQ}(\mathbf{p}) = \frac{1}{n'}\sum_{i=1}^{n'} K_{\mathbf{H}}\left(\mathbf{p}-\mathbf{d}'_i\right), \tag{2}$$

where $\mathbf{d}'_i \in Dataset'_{EQ}$ is a historical earthquake point from the augmented dataset. $K_H(\cdot)$ is a multidimensional Gaussian kernel function defined as:

$$K_{\mathbf{H}}(\mathbf{u}) = \frac{1}{2\pi|\mathbf{H}|^{1/2}}\exp\left(-\frac{1}{2}\mathbf{u}^T\mathbf{H}^{-1}\mathbf{u}\right), \tag{3}$$

where $\mathbf{u}=\mathbf{p}-\mathbf{d}_i'$ is the vector difference between the estimation point and the data point. $\mathbf{H}$ is the 2×2 Bandwidth Matrix, which controls the smoothness of the resulting density field.

We employ an automatic bandwidth selection method, such as Scott's rule or Silverman's rule, as a baseline. To better capture the fine-grained structures of seismic zones, we introduce an adjustment coefficient C_{adj}. The final bandwidth h is related to the automatically calculated bandwidth h_{auto} by $h=h_{auto}/C_{adj}$. By setting $C_{adj}>1$ (e.g., 1.6), we effectively reduce the bandwidth, resulting in a more detailed density map that better highlights the contours of major seismic belts.

The final seismic hazard probability model, as shown in Fig. 1, is formally defined as:

$$P_{EQ}(x, y) = \mathrm{KDE}_{\mathrm{Gaussian}}\left(Dataset'_{EQ}, \mathbf{H}\right). \tag{4}$$

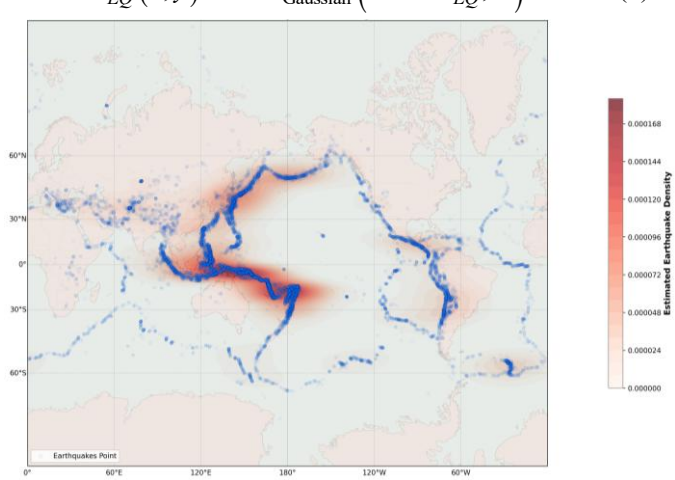

Fig. 1. Global Earthquake Probability Density

This model takes any longitude-latitude coordinate pair (x, y) as input and outputs a non-negative scalar value representing the probability density of an earthquake at that location. This output serves as a primary objective function in the subsequent multi-objective optimization.

D. Tropical Cyclone Hazard Probability Model

Analogous to the seismic model, this model aims to construct a spatial probability density function, $P_{TC}(x, y)$ to quantify the relative risk of encountering a tropical cyclone at any global location (x, y). The model is built upon historical data of complete tropical cyclone tracks, designed to identify high-frequency regions and primary storm paths [22-25]. The path points of tropical cyclones also exhibit a highly clustered and irregular spatial distribution, concentrated primarily over specific low-latitude ocean basins. Consequently, the KDE method is again the ideal choice for this modeling task. By applying KDE to the discrete points along all historical tracks, we can generate a smooth, continuous probability density field that accurately reflects tropical cyclone activity hotspots.

Let the historical tropical cyclone track point dataset be $Dataset_{TC} = \{\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_m\}$, where m is the total number of track points sampled from all historical records. Each data point $\mathbf{d}_j=(x_j, y_j)$ represents the longitude and latitude coordinates of a cyclone's center at a specific time. To ensure the generation of a continuous probability distribution across the Pacific Ocean's dateline, we employ the exact same data augmentation strategy as in the seismic model. This involves creating longitudinal-shifted copies of track points near the ±180° meridians to form the augmented dataset $Dataset'_{TC}$.

Based on the augmented track point dataset $Dataset'_{TC}$ (of size m′), the probability density of encountering a tropical cyclone at any location $\mathbf{p}=(x, y)$ is given by:

$$P_{TC}(\mathbf{p}) = \frac{1}{m'} \sum_{j=1}^{m'} K_{\mathbf{H}}\left(\mathbf{p} - \mathbf{d}'_j\right), \tag{5}$$

where the terms are defined identically to the seismic model, with the data points $\mathbf{d}'_j$ now originating from $Dataset'_{TC}$.

The tropical cyclone hazard probability model, as shown in Fig. 2, is formalized as:

$$P_{TC}(x, y) = \mathrm{KDE}_{\mathrm{Gaussian}}\left(Dataset'_{TC}, \mathbf{H}\right). \tag{6}$$

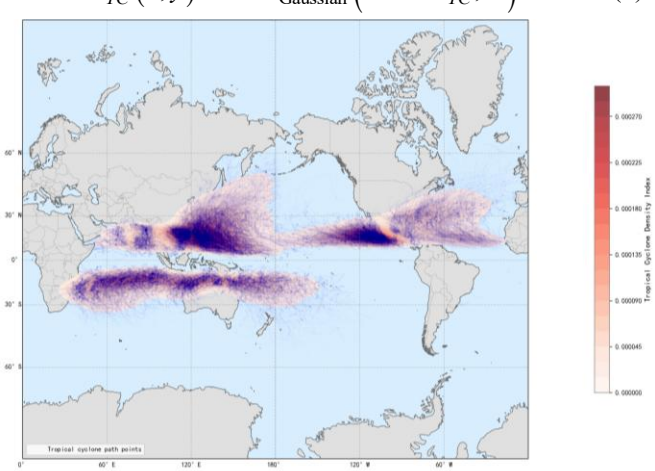

Fig. 2. Global Tropical Cyclone Probability Density

This model provides the second core objective function for the buoy layout optimization. By maximizing $P_{TC}(x, y)$, buoy is preferentially placed in the most historically active ocean regions to capture the most valuable data for early warnings.

E. Buoy Spatial Influence and Coverage Density Model

This model aims to quantify and evaluate the spatial uniformity of a given buoy network, which includes both existing and candidate buoys. To achieve this, we define a "coverage density function", $D_{Coverage}(x, y)$. The value of this function at any geographical point (x, y) represents the cumulative influence exerted on that point by all buoys in the network. The goal is to find a layout that minimizes high-density clusters and fills low-density gaps, thereby promoting a uniform monitoring presence. We model each buoy as a source of influence that radiates outwards from its center. This influence decays with distance, vanishing beyond a maximum influence radius, R_{max}. The total coverage density at any point in the ocean is defined as the linear superposition of the influence contributions from all individual buoys.

Let the set of buoy locations be $Dataset_{Buoy} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\}$, where k is the total number of buoys. Each point $\mathbf{b}_i=(x_i, y_i)$ is a vector of the i-th buoy's longitude and latitude. To accurately compute the distance between a point $\mathbf{p}=(x, y)$ and a buoy $\mathbf{b}_i$ on the Earth's surface, we use the Haversine formula. This formula calculates the great-circle distance $d(\mathbf{p},\mathbf{b}_i)$ between two points specified by their longitude and latitude.

The influence of a single buoy $\mathbf{b}_i$ on a point $\mathbf{p}$ is described by an influence function $I(\mathbf{p}, \mathbf{b}_i)$. This is a piecewise function dependent on the distance $d(\mathbf{p}, \mathbf{b}_i)$, with the influence normalized to 1 at the buoy's exact location (d=0). We primarily consider a linear decay model [26]:

$$I_{\mathrm{linear}}(\mathbf{p}, \mathbf{b}_i) = \begin{cases} 1 - \dfrac{d(\mathbf{p}, \mathbf{b}_i)}{R_{max}} & \text{if } d(\mathbf{p}, \mathbf{b}_i) \le R_{max} \\ 0 & \text{if } d(\mathbf{p}, \mathbf{b}_i) > R_{max} \end{cases}, \tag{7}$$

where R_{max} is the predefined maximum influence radius (e.g., 1000 km).

The total coverage density $D_{Coverage}(\mathbf{p})$ at any point $\mathbf{p}=(x, y)$ is the sum of influences from all k buoys in the network:

$$D_{Coverage}(\mathbf{p}) = \sum_{i=1}^{k} I(\mathbf{p}, \mathbf{b}_i). \tag{8}$$

The resulting function, $D_{Coverage}(x, y)$ as shown in Fig. 3, provides a density value for every point in the ocean. In the multi-objective optimization framework, the objective function f_3 associated with avoiding dense areas, to promote global uniformity.

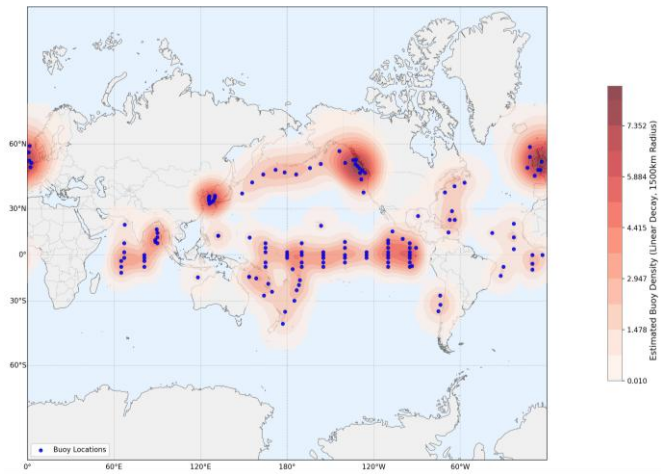

Fig. 3. Spatial Distribution of Existing Buoy

F. Geospatial Constraint Model

The objective of this model is to formally define the Geospatial Feasible Domain for buoy deployment. Not all marine locations are suitable for mooring buoy. This model serves to exclude infeasible regions based on two primary physical constraints. Firstly, we proposed the land constraint, as the buoy must be deployed in the ocean, not on land. Secondly, we proposed the water depth constraint, as buoy must be moored in water of sufficient depth to prevent grounding and to ensure that collected data are representative of open-ocean conditions. To facilitate efficient handling by optimization algorithms, we construct a continuous constraint violation function, C(x, y). This function assigns a non-negative "violation score" to any coordinate (x, y). A score of 0 indicates that all constraints are met, that is, the feasible location, while a score greater than 0 signifies a constraint violation, with the magnitude of the score proportional to the severity of the violation.

The construction of the violation function is based on two high-resolution global datasets. Firstly, vector data from Natural Earth was employed to define a precise multi-polygon object ($\mathbb{L}$) delineating all global landmasses. Secondly, the ETOPO1 global relief model [20, 27], which contains elevation and bathymetry data, was used to build a continuous bilinear interpolation function, $E(x, y)$. This function provides the elevation at any coordinate, where positive values indicate land and negative values indicate ocean depth.

Parameter definitions are as follows:

$\mathbb{L}$: The geometric set of all global landmasses.

$E(\mathbf{p})$: The elevation function at point $\mathbf{p}=(x, y)$. The water depth is thus $d_w(\mathbf{p})=-E(\mathbf{p})$.

d_{min}: The minimum water depth threshold for deployment (e.g., 100 meters).

C_{land_base}: A large base penalty for deploying on land (e.g., 1000).

C_{depth_base}: A penalty for deploying in shallow water (e.g., 500).

λ_{land}: A penalty scalar proportional to the distance inland (e.g., 1.0/km).

$d_{coast}(\mathbf{p})$: The shortest great-circle distance from an inland point $\mathbf{p}$ to the nearest coastline (the boundary of $\mathbb{L}$).

The violation function $C(\mathbf{p})$ is defined as a piecewise function based on a point's location relative to land and its water depth:

$$C(\mathbf{p})=\begin{cases} C_{land_base}+\lambda_{land}\cdot d_{coast}(\mathbf{p}) & \text{if } \mathbf{p}\in\mathbb{L} \\ C_{depth_base} & \text{if } \mathbf{p}\notin\mathbb{L}\wedge d_w(\mathbf{p})\le d_{min} \\ 0 & \text{if } \mathbf{p}\notin\mathbb{L}\wedge d_w(\mathbf{p})> d_{min} \end{cases}. \quad (9)$$

The constraint model is the function $C(x, y)$. Within the optimization algorithm, the total constraint violation V for a candidate solution with N_{buoy} buoys at locations $\{\mathbf{p}_1,\ldots,\mathbf{p}_{N_{buoy}}\}$ is the sum of individual violations:

$$V=\sum_{i=1}^{N_{buoy}} C(\mathbf{p}_i). \quad (10)$$

The optimization algorithm seeks solutions that minimize the objective functions while strictly satisfying the condition V=0. The Constraint Domination principle is used, where any feasible solution (V=0) is always preferred over any infeasible solution (V>0) [28]. This mechanism powerfully guides the search towards the feasible domain, ensuring that the final optimized layouts are physically deployable, as shown in Fig. 4.

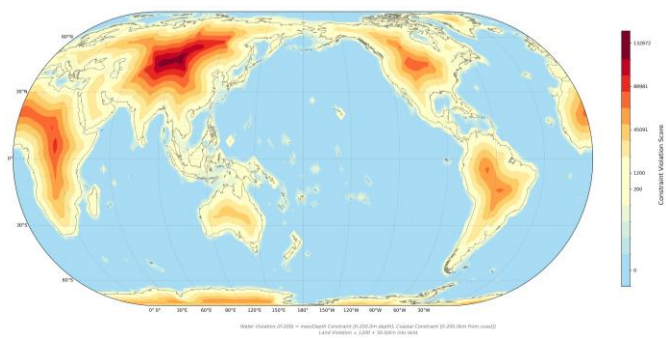

Fig. 4. Global Buoy Placement Constraint Violation Map

In summary, the moored ocean buoy layout optimization problem can be formulated as a CMMOP. The goal is to simultaneously optimize objective functions f_1 (seismic coverage), f_2 (tropical cyclone coverage), and f_3 (spatial uniformity), subject to constraint g_1. The extreme complexity of the probability models and constraints gives this problem its characteristic constrained, multimodal, and multi-objective nature.

G. Analysis of Model Properties: Non-linearity, Non-differentiability, and Non-convexity

A profound understanding of the model's inherent mathematical properties serves as the theoretical cornerstone for selecting an appropriate optimization algorithm. This section aims to rigorously demonstrate, from a mathematical standpoint, that the buoy layout optimization model we have constructed is intrinsically non-linear, non-convex, and that its constraint boundaries are non-differentiable. These properties collectively render traditional mathematical programming methods ineffective, thereby highlighting the necessity of employing an advanced intelligent optimization algorithm such as HACMMO.

The model's complexity primarily stems from two aspects: the objective functions (f_1, f_2) and the constraint function (g_1).

First, the objective functions f_1 and f_2, which are constructed based on KDE by superimposing a multitude of Gaussian kernel functions, are inherently non-linear. Due to the clustered distribution of real-world hazard data (e.g., the Pacific Ring of Fire), the resulting probability density surfaces inevitably form multiple distinct peaks. By definition, any function possessing multiple local optima is non-convex.

The core challenge, however, originates from the constraint function g_1, which defines the problem's feasible domain—the global ocean. This domain is a black box defined by Earth's true geographical data, for which no analytical expression is available for direct mathematical derivation.

To formally characterize the properties of this black-box model, we must introduce mathematical archetypes that can axiomatically capture its essential features. Fortunately, such archetypes are well-established and their validity has been extensively confirmed:

Archetype for the Coastline Boundary—The Koch Curve [29].

Archetype for Natural Surfaces like Seabed Topography—The Weierstrass-Mandelbrot (W-M) Function [30].

The selection of these two archetypes as our theoretical foundation is justified by their widespread acceptance and validation as mathematical models for complex natural phenomena. This foundation was laid by Benoit Mandelbrot [30], the "father of fractal geometry", in his pioneering work.

In his seminal paper, "How Long Is the Coast of Great Britain?" [31], Mandelbrot first established the profound that coastline is essentially a fractal, demonstrating that natural geographical boundaries possess fractal characteristics—they are continuous yet nowhere differentiable.

In his authoritative book, "The Fractal Geometry of Nature" [30], Mandelbrot further popularized the W-M function as the ideal mathematical model for describing the fractal phenomena ubiquitous in nature, such as terrain, clouds, and turbulence.

Subsequent research has consistently affirmed the efficacy of the W-M function in simulating complex terrains. For instance, Dawkins et al. [32, 33] have repeatedly employed the W-M function to generate fractal terrain surfaces for vehicle dynamics simulations. Similarly, Cui et al. [34] applied the W-M function to produce simulated digital terrain maps.

Based on this solid theoretical support, we can posit the following modeling premise.

Assumption 1. The key mathematical properties of our data-driven, black-box model that challenge traditional optimization algorithms can be faithfully and axiomatically characterized by their corresponding mathematical archetypes.

(1) The feasible solution space, whose geometric complexity is dictated by Earth's coastlines, is axiomatically characterized by the region defined by the Koch curve.

(2) The function surfaces, seabed topography, which are defined by rugged, data-driven landscapes, are axiomatically characterized by the properties of the W-M function.

Consequently, by analyzing the properties of these mathematical archetypes, we can deduce the intrinsic properties of our optimization model.

Theorem 1. *The region defined by the Koch snowflake is a non-convex set.*

Proof. A set K is convex if and only if for any two points A, $B \in K$, the line segment L connecting them is entirely contained within K. The line segment, as shown in Fig. 5, is defined as:

$$L = \{\lambda A + (1-\lambda)B \mid \lambda \in [0,1]\}. \tag{11}$$

To prove that the Koch region, K, is non-convex, we use proof by counterexample. Let us select a point A within the interior of one "arm" of the snowflake and a point B within the interior of a non-adjacent "arm", such that $A \in K$ and $B \in K$. The line segment AB must pass through the empty space between these arms, which lies outside of K. Therefore, a point on the segment, for instance the midpoint $M = 0.5A + 0.5B$, is not in K. This directly violates the definition of a convex set.

Therefore, the Koch region K is a non-convex set. This conclusion can be directly extended to the global ocean, a straight line connecting a feasible point in the Pacific Ocean to a feasible point in the Atlantic Ocean would necessarily pass through American continent. Thus, the global ocean is also a non-convex set. ■

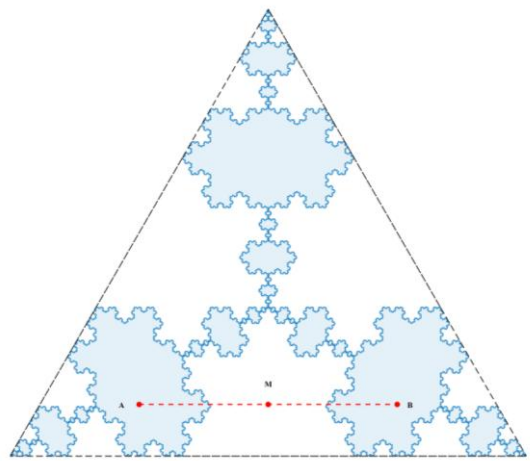

Fig. 5. Koch curve

To facilitate the proof, we adopt the Weierstrass-Mandelbrot (W-M) function, as shown in Fig. 6, in the following form and state the conditions it satisfies:

$$f(x) = C\sum_{n=0}^{\infty} b^{n(H-1)} \cos\left(2\pi b^n x + \phi_n\right), \tag{12}$$

where C is a positive scaling constant, $b>1$, $0<H<1$ is the Hurst exponent, and $b^{(H-1)}<1$. The φ_n are random phases.

Theorem 2. *The W-M function f(x) is non-linear.*

Proof. We proceed by contradiction. A linear function, such as $g(x)=mx+c$, has a second derivative that is identically 0, $g'(x)=0$. If we assume $f(x)$ is linear, its second derivative must also be 0. Assuming that term-by-term differentiation is permissible, we differentiate $f(x)$ twice:

$$f''(x) = C\sum_{n=0}^{\infty} b^{n(H-1)}\left[-\left(2\pi b^n\right)^2 \cos\left(2\pi b^n x + \phi_n\right)\right]$$
$$= -4C\pi^2 \sum_{n=0}^{\infty} b^{n(H-1)} b^{2n} \cos\left(2\pi b^n x + \phi_n\right) \quad . \quad (13)$$
$$= -4C\pi^2 \sum_{n=0}^{\infty} b^{n(H+1)} \cos\left(2\pi b^n x + \phi_n\right)$$

For $f(x)$ to be a linear function, its second derivative must be 0 for all x. However, since $b>1$ and $H>0$, it follows that $H+1>1$. This means the amplitude of the terms in the series, $b^{n(H+1)}$, grows exponentially as n increases. A series composed of an infinite number of cosine functions with different frequencies and exponentially diverging amplitudes cannot sum to 0 for all x.

This contradicts the necessary condition for a linear function, $f''(x)=0$. Therefore, the W-M function $f(x)$ is non-linear. ■

Theorem 3. *The W-M function f(x) possesses infinitely many local minima.*

Proof. The proof of this theorem hinges on the W-M function's satisfaction of the Reverse Hölder condition, a property thoroughly demonstrated by Zaleski [35]. This condition fundamentally ensures that the function is non-monotonic within any arbitrarily small interval, meaning it must exhibit continuous "undulations" at all scales [35]. As the W-M function is continuous, the Extreme Value Theorem guarantees that each of these necessary "undulations" within any small closed interval must contain at least one local maximum and one local minimum. Given that an infinite number of such small intervals can be found across the function's entire domain, it can be directly concluded that the number of its local extrema is infinite.

In conclusion, it is the Reverse Hölder condition that ensures the function's non-monotonicity in any small interval. This property, combined with continuity, necessarily leads to the existence of infinitely many local extrema. ■

Theorem 4. *Following from Theorem 3, the W-M function f(x) is non-convex.*

Lemma. If a continuous function $g(x)$ defined on the real number line $\mathbb{R}$ possesses at least two distinct strict local minima, then $g(x)$ is non-convex.

Proof of Lemma. Let x_a and x_e be two distinct strict local minima of $g(x)$, with $x_a<x_e$. By the Extreme Value Theorem, $g(x)$ must attain a maximum value on the closed interval $[x_a, x_e]$ at some point x_m. Since x_a and x_e are strict local minima, x_m must be in the open interval (x_a, x_e), which implies:

$$g(x_m) > max\left(g(x_a),\ g(x_e)\right). \quad (14)$$

The point x_m can be expressed as a convex combination of x_a and x_e,

$$x_m = \lambda x_e + (1-\lambda) x_a, \quad (15)$$

where $\lambda=(x_m-x_a)/(x_e-x_a)\in(0,\ 1)$. The definition of a convex function requires :

$$g(x_m) \le \lambda g(x_e) + (1-\lambda) g(x_a). \quad (16)$$

However, we have:

$$\begin{aligned} g(x_m) &> max\left(g(x_a), g(x_e)\right) \\ &\ge \lambda g(x_e) + (1-\lambda) g(x_a) \end{aligned}, \quad (17)$$

which is a direct contradiction. Thus, $g(x)$ must be non-convex. ■

Proof of Theorem 4. The W-M function $f(x)$ is the sum of a uniformly convergent series of continuous functions, and therefore it is itself continuous. According to Theorem 2, $f(x)$ possesses infinitely many strict local minima, and thus at least two. Applying the lemma above, we directly conclude that the W-M function $f(x)$ is non-convex. ■

Property 1. *The W-M function f(x) is nowhere differentiable.*

This is the most famous property of the function. Hardy [36] has proved that although the function's curve is continuous, it is impossible to define a unique tangent at any point.

The core idea is that for any point on the curve, one can always construct a sequence of nearby points for which the limit of the difference quotient (the slope) diverges to infinity. This behavior occurs as long as the parameters satisfy the condition $b^H \ge 1$. The violent oscillations of the high-frequency components are always present and prevent the slope from converging to a finite value.

A direct implication of this property is that any objective function or constraint modeled with the W-M function cannot be solved using gradient-based optimization methods. Standard local optimization algorithms are highly susceptible to becoming trapped in one of its infinitely many local optima.

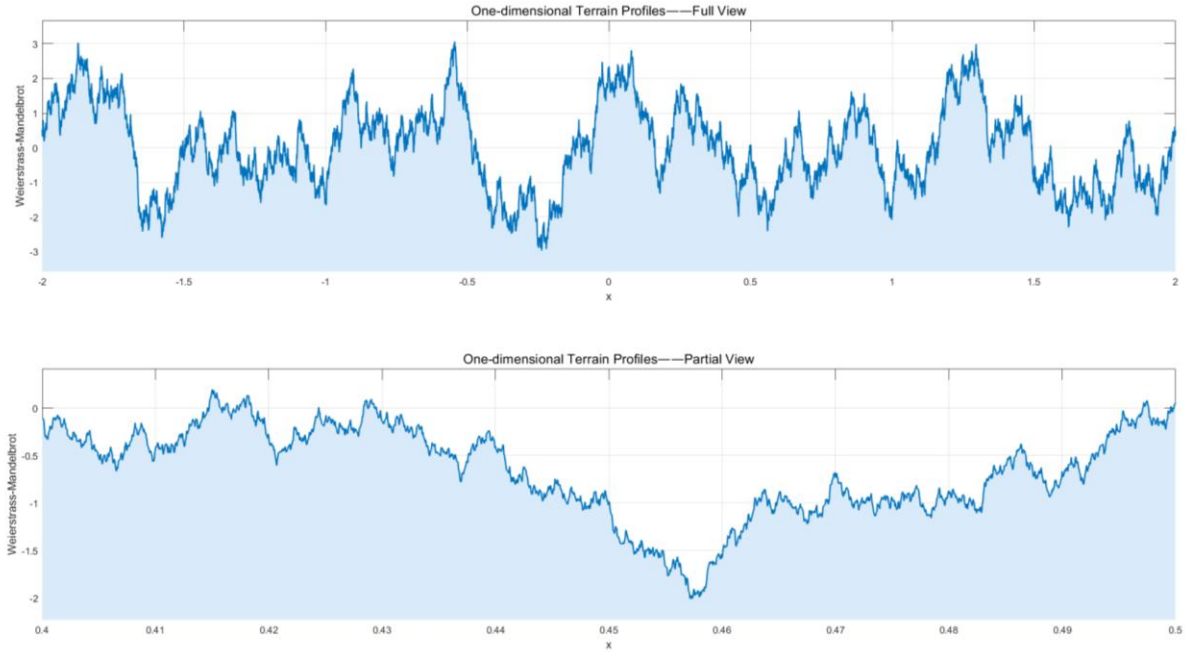

Fig. 6. (W-M) function——simulated one-dimensional terrain profiles

Thus, our analysis culminates in a clear understanding of the model's fundamental mathematical nature. We have rigorously demonstrated that both the objective and constraint functions are highly non-linear and non-convex. Moreover, the feasible domain itself—the geometric shape of the global oceans—is inherently non-convex. Most critically for the selection of a solver, the constraint boundaries exhibit fractal properties; while continuous, they are also nowhere differentiable. This complete absence of gradient information renders traditional gradient-based optimization methods inapplicable and underscores the necessity for an advanced, gradient-free algorithmic approach.

IV. A Historical Archive-Assisted Co-evolutionary Algorithm for CMMOP

Traditional multi-objective optimization algorithms often struggle with complex constraints, easily becoming trapped in local optima and failing to find multiple high-quality Pareto optimal solutions across a vast decision space. To effectively solve the buoy layout problem, a typical constrained multimodal multi objective optimization problem (CMMOP), we propose a co-evolutionary algorithm assisted by a historical archive, hereafter referred to as HACMMO. Its core idea is to use two co-evolving populations, one focusing on feasible solutions and the other on unconstrained exploration and leveraging a de-duplicated historical archive to store and guide the search for global optimal. The modeling and solving framework is shown in Fig. 7.

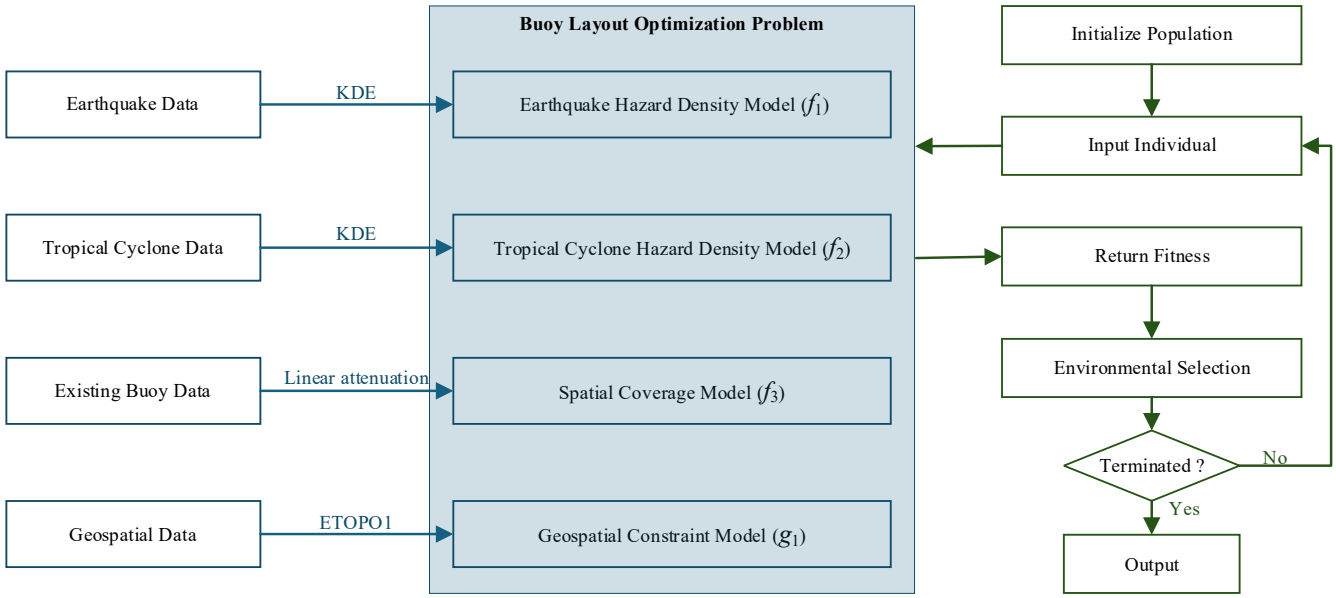

Fig. 7. Framework of the Buoy Layout Optimization System.

A. Algorithm Framework

The HACMMO framework consists of three core components: a constrained population, an unconstrained population, and a de-duplicated elite historical archive. Its procedure is outlined in Algorithm 1.

The constrained population, central to the algorithm, evolves strictly adhering to physical and geographical constraints, aiming to identify high-quality Pareto optimal solutions within the feasible region. Fitness evaluation follows the classic constraint-domination principle, solutions with smaller constraint violations are preferred, for solutions with equal violation levels, Pareto dominance is used for selection.

Conversely, the unconstrained population disregards all constraints, exploring the solution space solely based on objective function values which allows it to freely search the entire decision space to discover potentially superior objective function landscapes. This helps the constrained population escape local optima.

The de-duplicated historical archive stores all non-dominated solutions discovered by both the constrained and unconstrained populations. To ensure solution diversity and prevent redundancy, a strict de-duplication process is applied before adding a new solution: a solution is accepted only if its decision variables and objective function values are distinct from all existing solutions in the archive. This archive not only preserves the best solutions found so far but also provides valuable guidance for the evolution of populations.

During the parallel evolution of the constrained and unconstrained populations, an information exchange mechanism facilitates mutual benefit, while the historical archive continuously assimilates and maintains the diversity of optimal solutions.

Algorithm 1 HACMMO

Input: N (population size)
Output: *Archive*

1: $Pop_c \leftarrow$ Initialize constrained population of size N
2: $Pop_u \leftarrow$ Initialize unconstrained population of size N
3: $Archive \leftarrow$ Initialize elite archive (empty at beginning)
4: Evaluate fitness of Pop_c and Pop_u
5: **while** termination conditions not met **do**
6: $O_c \leftarrow$ Generate offspring from Pop_c using genetic operators
7: $O_u \leftarrow$ Generate offspring from Pop_u using genetic operators
8: $Pop_c \leftarrow$ Select N individuals from $[Pop_c \cup O_c \cup O_u]$ using SPEA2-based environmental selection with Constraint Domination Principle (CDP)
9: $Pop_u \leftarrow$ Select N individuals from $[Pop_c \cup O_c \cup O_u]$ using SPEA2-based environmental selection with Pareto Dominance
10: $Archive \leftarrow$ Update with de-duplicated, non-dominated solutions from $[Pop_c \cup Pop_u \cup Archive]$
11: **end while**
12: **return** *Archive*

B. Core Evolutionary Operators and Constraint Handling

Each individual is encoded as a real-valued vector representing the latitude and longitude coordinates of a set of new buoys. The initial population is generated randomly across the global domain. Fitness evaluation for both populations draws upon the strength Pareto evolutionary algorithm (SPEA2), incorporating individual strength (S), raw fitness (R), and density information (D). In the constrained population, constraint violations are addressed prior to computing R and D, whereas the unconstrained population derives fitness directly

from objective function values. If non-dominated solutions exceed the population size, the most crowded individuals are removed to preserve diversity. A key feature lies in the offspring generation and information exchange, where environmental selection for both populations considers parents and offspring from both populations, thereby fostering effective synergy.

V. Experiments and Results

To validate the effectiveness of the proposed HACMMO algorithm in solving the global ocean buoy layout optimization problem, we conducted a series of comparative experiments by using MATLAB [37, 38].

A.Experimental Setup

We directly applied the global ocean buoy layout optimization model constructed in the previous section. The objectives are to maximize earthquake and tropical cyclone monitoring probabilities while optimizing the uniformity of the buoy distribution, subject to water depth and coastal distance constraints.

We selected four SOTA CMMO algorithms for comparison: constrained multimodal multi-objective differential evolution (CMMODE) [39], a variation of the differential evolution algorithm; constrained multimodal multi-objective co-evolutionary algorithm (CMMOCEA) [28], a co-evolutionary algorithm; a method based on PSO known as constrained dominance principle, multi-objective, ring particle swarm optimization with special crowding distance (CDP_MO_Ring_PSO_SCD) [40]; and constrained dominance principle with decision space niching non-dominated sorting genetic algorithm II (CDP_DNNSGAII) [41], an advancement of the well-known NSGA-II.

As this is a real-world application, the true Pareto front and optimal solution set are unknown. However, metrics like inverted generational distance (IGD) [37], and inverted generational distance in decision space (IGDX) [37], which require a reference front, are not applicable. To solve this problem, we merge the final output populations from all runs of all algorithms and perform non-dominated sorting [42]. Then, the resulting non-dominated solutions are treated as the "true" Pareto front, which makes it possible to use IGD, and IGDX as metrics to evaluate the performance of each algorithm.

Specifically, we aggregated the final populations from 30 independent runs of each of the five algorithms, where each run terminated after 30,000 function evaluations (*maxFE* = 30,000). Since each run concluded with 100 individuals, this created a comprehensive pool of 15,000 solutions. Non-dominated sorting was then applied to this entire pool, from which 1,561 non-dominated solutions belonging to the first front were identified. Consequently, the objective vectors and decision vectors of these 1,561 solutions were designated as the reference "true" Pareto front (PF) and "true" Pareto set (PS), respectively.

Tables I, II, and III present the performance metrics hypervolume (HV), IGDX and IGD for HACMMO and four other algorithms on the problem.

The population size for all algorithms was set to 200. To eliminate random effects, each algorithm was run independently 30 times.

TABLE I
MEAN AND STANDARD DEVIATION OF HV FOR FIVE ALGORITHMS

Problem	maxFE	CMMODE	CMMOCEA	CDP_DNNSGAII	CDP_MO_Ring_PSO_SCD	HACMMO
Buoy Layout	10000	4.3561e-1(3.44e-3)	4.4109e-1(9.27e-3)	4.3062e-1(9.91e-3)	4.3176e-1 (3.17e-3)	**4.7290e-1 (1.34e-3)**
	20000	4.3652e-1 (3.90e-3)	4.4217e-1 (9.60e-3)	4.3523e-1 (1.46e-2)	4.3202e-1 (3.86e-3)	**4.7336e-1(1.78e-3)**
	30000	4.3639e-1(4.49e-3)	4.4835e-1(1.33e-2)	4.3501e-1(1.26e-2)	4.3371e-1(8.71e-3)	**4.7367e-1(5.77e-3)**
+/-/≈		0/3/0	0/3/0	0/3/0	0/3/0	

TABLE II
MEAN AND STANDARD DEVIATION OF IGD FOR FIVE ALGORITHMS

Problem	maxFE	CMMODE	CMMOCEA	CDP_DNNSGAII	CDP_MO_Ring_PSO_SCD	HACMMO
Buoy Layout	10000	3.8298e-2(2.20e-2)	3.0784e-2(3.88e-3)	3.3319e-2 (3.70e-3)	4.2495e-2(4.38e-3)	**2.8402e-2(3.32e-4)**
	20000	3.8737e-2 (2.15e-2)	6.9969e-2 (2.29e-1)	3.2135e-2(4.02e-3)	3.9062e-2 (3.61e-3)	**2.7775e-2 (8.67e-4)**
	30000	3.2634e-2 (1.56e-3)	2.8735e-2(2.69e-3)	3.2149e-2(3.09e-3)	3.7451e-2 (2.92e-3)	**2.7727e-2(1.58e-3)**
+/-/≈		0/3/0	0/3/0	0/3/0	0/3/0	

TABLE III
MEAN AND STANDARD DEVIATION OF IGDX FOR FIVE ALGORITHMS

Problem	maxFE	CMMODE	CMMOCEA	CDP_DNNSGAII	CDP_MO_Ring_PSO_SCD	HACMMO
Buoy Layout	10000	1.2807e+0 (1.44e+0)	3.4816e+0 (6.06e+0)	3.1476e+0(6.33e+0)	**1.2788e+0(1.59e-1)**	1.5791e+0(1.49e+0)
	20000	1.1416e+0 (6.83e-1)	1.2550e+0 (1.36e+0)	5.2718e+0 (1.02e+1)	1.0784e+0(8.26e-2)	**9.5986e-1(1.00e+0)**
	30000	1.2578e+0 (9.27e-1)	2.3749e+0(6.28e+0)	3.9443e+0(8.55e+0)	1.0442e+0 (8.68e-2)	**7.2847e-1(3.03e-2)**
+/-/≈		0/3/0	0/3/0	0/3/0	1/2/0	

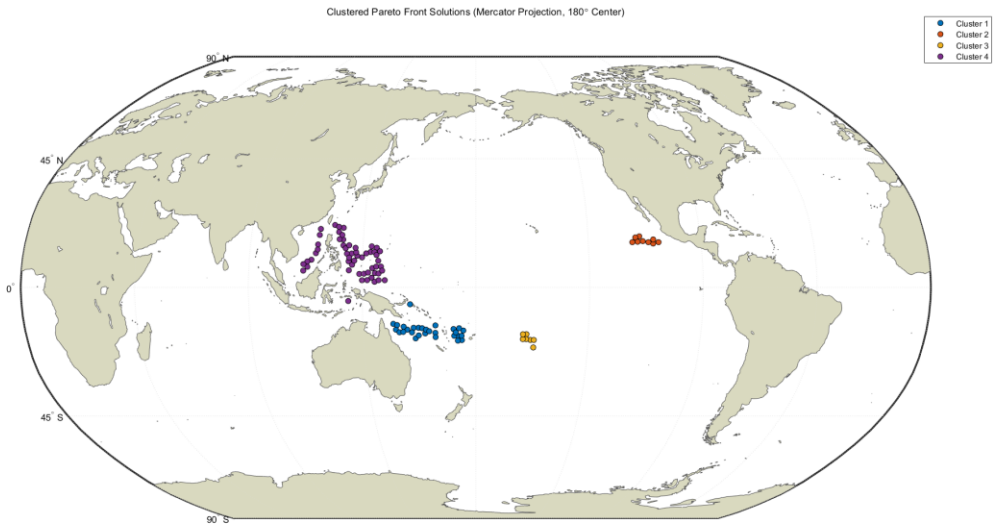

Fig. 8. Geographical Distribution of Pareto Optimal Solutions

B. Results and Discussion

Our comprehensive experimental results across the HV, IGD, and IGDX metrics, as presented in Tables I-III, robustly validate the superiority of the proposed HACMMO. Overall, HACMMO secured the best mean result in 8 out of 9 test scenarios, clearly demonstrating its dominant performance.

Specifically, the results for the HV metric in Table I are unequivocal, HACMMO surpassed all competitors across all *maxFE* settings, which indicates its exceptional ability to balance convergence and diversity in the objective space. This strength is mirrored by the IGD results in Table II, where HACMMO again achieved the top rank, confirming that its solution sets can best approximate the true Pareto front.

A more nuanced phenomenon is observed in the IGDX metric, as shown in Table III. At the lowest *maxFE* setting, HACMMO's performance was slightly outperformed by CDP_MO_Ring_PSO_SCD. This may suggest that the algorithm's search mechanism initially prioritizes convergence in the objective space, potentially at the expense of exploration quality in the decision space. Nevertheless, as the number of evaluations increases, HACMMO still achieves the best results on the IGDX metric. Therefore, while the current search strategy is highly effective, future work could still explore how to further balance the search efforts between the objective and decision spaces to achieve faster initial performance.

As shown in Fig. 8, the layout result of one of the runs of HACMMO demonstrates that the Pareto set is comprised of four distinct clusters (Clusters 1-4), which correspond to four subsets of solutions in the decision space. These subsets represent a Pareto-optimal trade-off among the three objectives: maximizing coverage of seismic and tropical cyclone zones while ensuring spatial separation from existing buoy arrays. Crucially, all solutions satisfy the required constraints for coastal distance and ocean depth, aligning with real-world buoy deployment requirements and thereby demonstrating the practical value of the proposed method.

In conclusion, the collective evidence from these comprehensive metrics jointly proves that, compared to other state-of-the-art algorithms, HACMMO can generate solutions of higher quality and with greater stability, making it a more outstanding algorithm. This superiority is primarily due to its unique co-evolutionary framework.

Firstly, the exploratory capability of the unconstrained population allows it to discover promising regions far from the current feasible solutions. This information is then passed to the constrained population, effectively helping the algorithm escape local optima caused by complex constraints.

Secondly, the de-duplicated elite historical archive not only preserves high-quality, diverse solutions but also provides a stable guide for the entire evolutionary process, accelerating convergence towards the global Pareto front.

Finally, through dual de-duplication of both decision variables and objective function values, along with consideration of crowding during environmental selection, HACMMO effectively maintains solution diversity, enabling it to find multiple, well-distributed layout schemes across the vast ocean space.

In summary, the outstanding performance of HACMMO demonstrates that it is a powerful and suitable tool for solving complex real-world buoy layout optimization problems.

V. Conclusion

This paper addresses the critical need to enhance global disaster warning capabilities by studying the optimization of moored ocean buoy layouts. For the first time, we systematically modeled this problem as a constrained multimodal multi-objective optimization problem, integrating historical earthquake and tropical cyclone probability data, buoy spatial uniformity, and multiple physical and geographical constraints. To solve this complex model, we proposed a co-evolutionary algorithm named HACMMO. This algorithm effectively balances global exploration and local exploitation through the search of constrained and unconstrained populations, guided by a de-duplicated elite historical archive, successfully overcoming the tendency of traditional algorithms to fall into local optima.

Experimental results show that, compared to other SOTA algorithms, our proposed HACMMO algorithm achieves a significant advantage on the HV indicator, demonstrating its ability to provide superior and more robust solutions for the buoy layout problem. This research not only provides a scientific decision-support tool for the design of global ocean observation networks but also demonstrates the effectiveness and potential of advanced intelligent optimization algorithms in solving major real-world challenges.

Future work can be extended in the following areas:

Integrating economic and engineering factors, such as buoy deployment and maintenance costs and data transmission bandwidth, as new optimization objectives for a more comprehensive multi-objective trade-off analysis.

Empowering other layout modeling areas, the buoy layout problem can be abstracted into a systematic workflow composed of data-driven approaches, black-box modeling, and optimization algorithms. This general framework is extensible to other layout optimization challenges. Potential practical applications include optimizing the placement of rain gauge stations, transmission towers, and signal base stations. The choice of the black-box modeling algorithm can be tailored to the specific characteristics of the problem and dataset, with potential candidates including KDE, Kriging surrogate model, Gaussian Mixture Models (GMM), eXtreme gradient boosting (XGBoost), and back propagation (BP) neural networks.

Ultimately, this work introduces a novel data-driven paradigm, presenting a new solution pathway as an alternative to traditional methods.

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